$$g_c = 386.09\ \frac{\text{lbm}\cdot\text{in}}{\text{lbf}\cdot\text{s}^2}, \qquad K_u = 1.20, \qquad W_p = 1.500\ \text{lbf}$$
$$m_p = 1.500\ \text{lbm}, \qquad \tau_{32,\text{cont}} = 88.5\ \text{lbf}\cdot\text{in}, \ \tau_{22,\text{cont}} = 32.6\ \text{lbf}\cdot\text{in}$$

Lower Shoulder Joint

$$W_{d,L} = 2.942\ \text{lbf}, \qquad m_{d,L} = 2.942\ \text{lbm}, \qquad x_{d,L} = 6.924\ \text{in}$$
$$d_{d,L} = 6.948\ \text{in}, \qquad I_{yy,G,L} = 55.165\ \text{lbm}\cdot\text{in}^2, \qquad x_{p,L} = 16.369\ \text{in}$$
$$d_{p,L} = 16.370\ \text{in}, \qquad \omega_{\max,L} = 0.6777\ \text{rad/s}, \qquad \alpha_{\max,L} = 2.4264\ \text{rad/s}^2\ (309\text{–}374)$$

$$\tau_{g,d,L} = W_{d,L}x_{d,L} = (2.942)(6.924) = 20.3704\ \text{lbf}\cdot\text{in}$$
$$\tau_{g,p,L} = W_p x_{p,L} = (1.500)(16.369) = 24.5535\ \text{lbf}\cdot\text{in}$$
$$\tau_{g,L} = \tau_{g,d,L} + \tau_{g,p,L} = 20.3704 + 24.5535 = 44.9239\ \text{lbf}\cdot\text{in}$$
$$I_{d,L} = I_{yy,G,L} + m_{d,L}d_{d,L}^2$$
$$= 55.165 + (2.942)(6.948)^2 = 197.1892\ \text{lbm}\cdot\text{in}^2$$
$$I_{p,L} = m_p d_{p,L}^2 = (1.500)(16.370)^2 = 401.9654\ \text{lbm}\cdot\text{in}^2$$
$$I_L = I_{d,L} + I_{p,L} = 197.1892 + 401.9654 = 599.1545\ \text{lbm}\cdot\text{in}^2$$
$$\tau_{I,L} = \frac{I_L\alpha_{\max,L}}{g_c} = \frac{(599.1545)(2.4264)}{386.09} = 3.7654\ \text{lbf}\cdot\text{in}$$
$$\tau_{\text{dyn},L} = \tau_{g,L} + \tau_{I,L} = 44.9239 + 3.7654 = 48.6893\ \text{lbf}\cdot\text{in}$$
$$\tau_{u,L} = K_u\tau_{\text{dyn},L} = (1.20)(48.6893) = \boxed{58.4272\ \text{lbf}\cdot\text{in}}$$
$$n_L = \frac{\tau_{32,\text{cont}}}{\tau_{u,L}} = \frac{88.5}{58.4272} = 1.5147 \approx \boxed{1.51}$$

Upper Shoulder Joint

$$W_{d,S} = 1.532\ \text{lbf}, \qquad m_{d,S} = 1.532\ \text{lbm}, \qquad W_{e,S} = 0.150\ \text{lbf}$$
$$m_{e,S} = 0.150\ \text{lbm}, \qquad d_{d,S} = d_{e,S} = 7.144\ \text{in}, \qquad I_{yy,G,S} = 12.617\ \text{lbm}\cdot\text{in}^2$$
$$d_{p,S} = 12.728\ \text{in}, \qquad \omega_{\max,S} = 0.6777\ \text{rad/s}, \qquad \alpha_{\max,S} = 2.4264\ \text{rad/s}^2\ (309\text{–}374)$$

$$\tau_{g,d,S} = W_{d,S}d_{d,S} = (1.532)(7.144) = 10.9446\ \text{lbf}\cdot\text{in}$$
$$\tau_{g,e,S} = W_{e,S}d_{e,S} = (0.150)(7.144) = 1.0716\ \text{lbf}\cdot\text{in}$$
$$\tau_{g,p,S} = W_p d_{p,S} = (1.500)(12.728) = 19.0920\ \text{lbf}\cdot\text{in}$$
$$\tau_{g,S} = 10.9446 + 1.0716 + 19.0920 = 31.1082\ \text{lbf}\cdot\text{in}$$
$$I_{d,S} = I_{yy,G,S} + m_{d,S}d_{d,S}^2$$
$$= 12.617 + (1.532)(7.144)^2 = 90.8053\ \text{lbm}\cdot\text{in}^2$$
$$I_{e,S} = m_{e,S}d_{e,S}^2 = (0.150)(7.144)^2 = 7.6555\ \text{lbm}\cdot\text{in}^2$$
$$I_{p,S} = m_p d_{p,S}^2 = (1.500)(12.728)^2 = 243.0030\ \text{lbm}\cdot\text{in}^2$$
$$I_S = I_{d,S} + I_{e,S} + I_{p,S}$$
$$= 90.8053 + 7.6555 + 243.0030 = 341.4638\ \text{lbm}\cdot\text{in}^2$$
$$\tau_{I,S} = \frac{I_S\alpha_{\max,S}}{g_c} = \frac{(341.4638)(2.4264)}{386.09} = 2.1459\ \text{lbf}\cdot\text{in}$$
$$\tau_{\text{dyn},S} = \tau_{g,S} + \tau_{I,S} = 31.1082 + 2.1459 = 33.2542\ \text{lbf}\cdot\text{in}$$
$$\tau_{u,S} = K_u\tau_{\text{dyn},S} = (1.20)(33.2542) = \boxed{39.9050\ \text{lbf}\cdot\text{in}}$$
$$n_S = \frac{\tau_{32,\text{cont}}}{\tau_{u,S}} = \frac{88.5}{39.9050} = 2.2178 \approx \boxed{2.22}$$

Elbow Joint

$$W_{d,E} = 0.892 \text{ lbf}, \qquad m_{d,E} = 0.892 \text{ lbm}, \qquad W_{e,E} = 0.100 \text{ lbf}$$
$$m_{e,E} = 0.100 \text{ lbm}, \qquad d_{d,E} = d_{e,E} = 4.700 \text{ in}, \qquad I_{yy,G,E} = 2.261 \text{ lbm·in}^2$$
$$d_{p,E} = 8.420 \text{ in}, \qquad \omega_{\max,E} = 0.5919 \text{ rad/s}, \qquad \alpha_{\max,E} = 1.1643 \text{ rad/s}^2 \text{ (0–149)}$$

$$\tau_{g,d,E} = W_{d,E} d_{d,E} = (0.892)(4.700) = 4.1924 \text{ lbf·in}$$
$$\tau_{g,e,E} = W_{e,E} d_{e,E} = (0.100)(4.700) = 0.4700 \text{ lbf·in}$$
$$\tau_{g,p,E} = W_p d_{p,E} = (1.500)(8.420) = 12.6300 \text{ lbf·in}$$
$$\tau_{g,E} = 4.1924 + 0.4700 + 12.6300 = 17.2924 \text{ lbf·in}$$
$$I_{d,E} = I_{yy,G,E} + m_{d,E} d_{d,E}^2$$
$$= 2.261 + (0.892)(4.700)^2 = 21.9653 \text{ lbm·in}^2$$
$$I_{e,E} = m_{e,E} d_{e,E}^2 = (0.100)(4.700)^2 = 2.2090 \text{ lbm·in}^2$$
$$I_{p,E} = m_p d_{p,E}^2 = (1.500)(8.420)^2 = 106.3446 \text{ lbm·in}^2$$
$$I_E = I_{d,E} + I_{e,E} + I_{p,E}$$
$$= 21.9653 + 2.2090 + 106.3446 = 130.5189 \text{ lbm·in}^2$$
$$\tau_{I,E} = \frac{I_E \alpha_{\max,E}}{g_c} = \frac{(130.5189)(1.1643)}{386.09} = 0.3936 \text{ lbf·in}$$
$$\tau_{\text{dyn},E} = \tau_{g,E} + \tau_{I,E} = 17.2924 + 0.3936 = 17.6860 \text{ lbf·in}$$
$$\tau_{u,E} = K_u \tau_{\text{dyn},E} = (1.20)(17.6860) = \boxed{21.2232 \text{ lbf·in}}$$
$$n_E = \frac{\tau_{22,\text{cont}}}{\tau_{u,E}} = \frac{32.6}{21.2232} = 1.5361 \approx \boxed{1.54}$$

Wrist Joint

$$W_{d,W} = 0.252 \text{ lbf}, \qquad m_{d,W} = 0.252 \text{ lbm}, \qquad W_{e,W} = 0.050 \text{ lbf}$$
$$m_{e,W} = 0.050 \text{ lbm}, \qquad d_{d,W} = d_{e,W} = 1.773 \text{ in}, \qquad I_{yy,G,W} = 0.862 \text{ lbm·in}^2$$
$$d_{p,W} = 3.926 \text{ in}, \qquad \omega_{\max,W} = 0.1639 \text{ rad/s}, \qquad \alpha_{\max,W} = 0.3365 \text{ rad/s}^2 \text{ (154–269)}$$

$$\tau_{g,d,W} = W_{d,W} d_{d,W} = (0.252)(1.773) = 0.4468 \text{ lbf·in}$$
$$\tau_{g,e,W} = W_{e,W} d_{e,W} = (0.050)(1.773) = 0.0887 \text{ lbf·in}$$
$$\tau_{g,p,W} = W_p d_{p,W} = (1.500)(3.926) = 5.8890 \text{ lbf·in}$$
$$\tau_{g,W} = 0.4468 + 0.0887 + 5.8890 = 6.4244 \text{ lbf·in}$$
$$I_{d,W} = I_{yy,G,W} + m_{d,W} d_{d,W}^2$$
$$= 0.862 + (0.252)(1.773)^2 = 1.6542 \text{ lbm·in}^2$$
$$I_{e,W} = m_{e,W} d_{e,W}^2 = (0.050)(1.773)^2 = 0.1572 \text{ lbm·in}^2$$
$$I_{p,W} = m_p d_{p,W}^2 = (1.500)(3.926)^2 = 23.1202 \text{ lbm·in}^2$$
$$I_W = I_{d,W} + I_{e,W} + I_{p,W}$$
$$= 1.6542 + 0.1572 + 23.1202 = 24.9316 \text{ lbm·in}^2$$
$$\tau_{I,W} = \frac{I_W \alpha_{\max,W}}{g_c} = \frac{(24.9316)(0.3365)}{386.09} = 0.0217 \text{ lbf·in}$$
$$\tau_{\text{dyn},W} = \tau_{g,W} + \tau_{I,W} = 6.4244 + 0.0217 = 6.4462 \text{ lbf·in}$$
$$\tau_{u,W} = K_u \tau_{\text{dyn},W} = (1.20)(6.4462) = \boxed{7.7354 \text{ lbf·in}}$$
$$n_W = \frac{\tau_{22,\text{cont}}}{\tau_{u,W}} = \frac{32.6}{7.7354} = 4.2144 \approx \boxed{4.21}$$

Final Continuous-Torque Margins

$$n_L = \boxed{1.51}, \qquad n_S = \boxed{2.22}, \qquad n_E = \boxed{1.54}, \qquad n_W = \boxed{4.21}$$

$\tau_{\text{motor,cont}} > K_u (\tau_g + \tau_I)$ for every analyzed arm joint.

Torque magnitudes are shown. The maximum dynamic case conservatively combines maximum gravity torque with peak angular acceleration. The same local-Y shoulder motion channel is applied to both shoulder actuator checks. The lower-shoulder value $m_{d,L} = 2.942$ lbm is treated as the complete selected downstream rotating assembly.